Experiment 6: POS Tagging and Named Entity Recognition

A. POS Tagging (Parts of Speech Tagging)

```
import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
True
```

```
from nltk.tokenize import word_tokenize
from nltk import pos_tag

text = "Natural Language Processing enables computers to understand l

tokens = word_tokenize(text)
tagged_words = pos_tag(tokens)

print("Input Text:", text)
print("\nWord and POS Tags:")

for word, tag in tagged_words:
    print(f"{word:15s} -> {tag}")
```

```
Input Text: Natural Language Processing enables computers to understand langu

Word and POS Tags:
Natural         -> JJ
Language        -> NNP
Processing      -> NNP
enables         -> VBZ
computers       -> NNS
to              -> TO
understand      -> VB
language        -> NN
.               -> .
```

B. Named Entity Recognition (NER)

```
import nltk
from nltk import word_tokenize, pos_tag, ne_chunk

nltk.download('maxent_ne_chunker_tab')
nltk.download('words')

text = "Elon Musk founded SpaceX in the United States."

# Tokenize the input
tokens = word_tokenize(text)

# Assign POS tags
tagged_words = pos_tag(tokens)

# Identify named entities
named_entities = ne_chunk(tagged_words, binary=False)

print("Named Entity Recognition Result:")
print(named_entities)

print("\nExtracted Entities:")

for entity in named_entities:
    if hasattr(entity, "label"):
        entity_name = " ".join(word for word, tag in entity.leaves())
        print(f"{entity_name} -> {entity.label()}")
```

```
[nltk_data] Downloading package maxent_ne_chunker_tab to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping chunkers/maxent_ne_chunker_tab.zip.
[nltk_data] Downloading package words to /root/nltk_data...
[nltk_data]   Unzipping corpora/words.zip.
Named Entity Recognition Result:
(S
  (PERSON Elon/NNP)
  (PERSON Musk/NNP)
  founded/VBD
  (ORGANIZATION SpaceX/NNP)
  in/IN
  the/DT
  (GPE United/NNP States/NNPS)
  ./.)

Extracted Entities:
Elon -> PERSON
Musk -> PERSON
SpaceX -> ORGANIZATION
United States -> GPE
```